Ex-10
8/10/2024

INTERNET WORKING

10) a)

INTER Networking USING ROUTER

AIM:-

To design and configure a simple internetwork using a router in Cisco Packet Tracer and verify the connectivity between two PCs.

OBSERVATION:-

1) A router and two PCs are used to create a simple internetwork.

2) PC0 is connected to the first interface of Router 1 using a copper straight-through cable.

3) PC1 is connected to the second interface of Router 1.

4) Router interfaces are assigned different network addresses.
   → 192.168.10.1/24
   → 192.168.20.1/24

5) PC0 is assigned IP address 192.168.10.2 with gateway 192.168.10.1

6) PC1 is assigned IP address 192.168.20.2 with gateway 192.168.20.1

7) The router forwards packets between the two different networks.

8) A PDU is sent from PC0 to PC1 to verify network connectivity.

```
Router>en
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#do show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         unassigned      YES unset  administratively down down
GigabitEthernet0/1         unassigned      YES unset  administratively down down
GigabitEthernet0/2         unassigned      YES unset  administratively down down
Vlan1                      unassigned      YES unset  administratively down down

Router(config)#int gigabitEthernet0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int g0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end



Router#show ip interface brief
Interface                  IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0         10.0.0.1        YES manual up                    up
GigabitEthernet0/1         192.168.1.1     YES manual up                    up
GigabitEthernet0/2         unassigned      YES unset  administratively down down
Vlan1                      unassigned      YES unset  administratively down down
Router#copy running-config sta-config
                           ^
% Invalid input detected at '^' marker.

Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#
```

```
!
!
spanning-tree mode pvst
!
!
!
!
!
interface GigabitEthernet0/0
 ip address 10.0.0.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
!
ip flow-export version 9
!
!
```

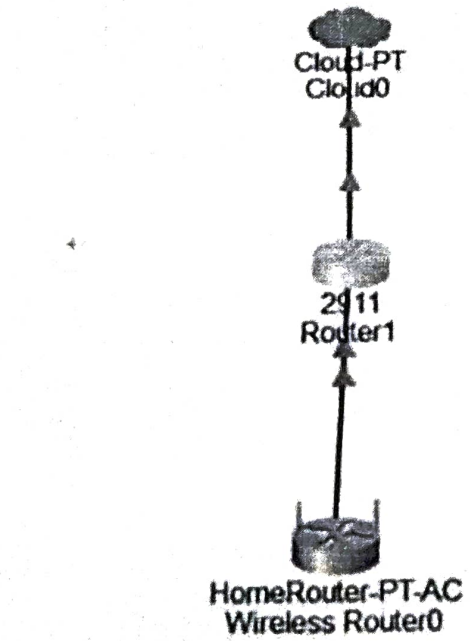
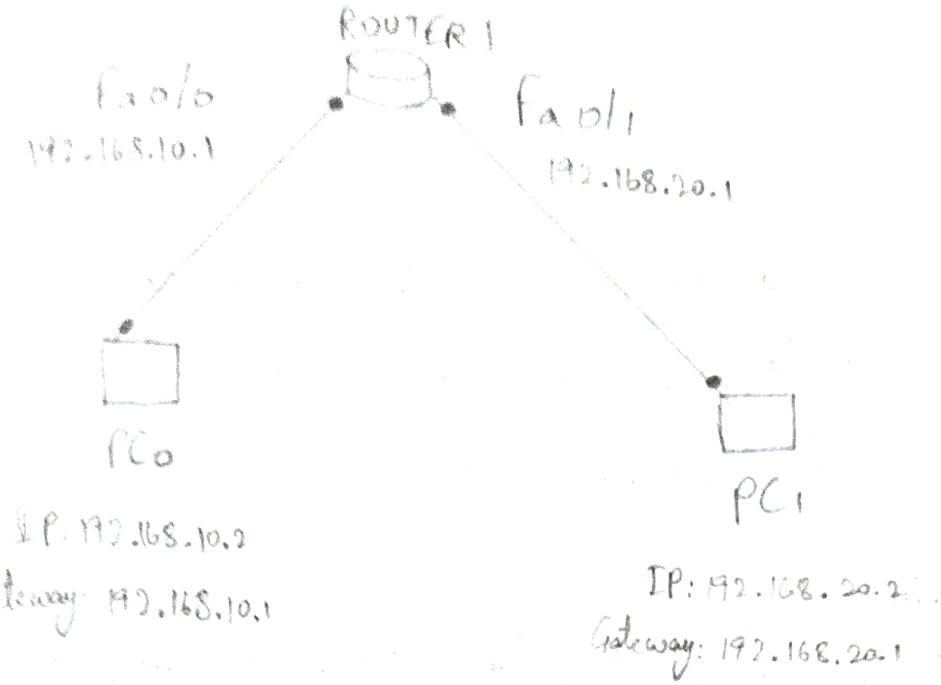




DIAGRAM:-



RESULT:-

Thus a simple * internetwork was successfully designed and configured using a router, and communication between PC0 and PC1 was successfully verified.

10) b)
8/10/2026

WIRELESS ROUTER, DHCP SERVER AND INTERNET CLOUD

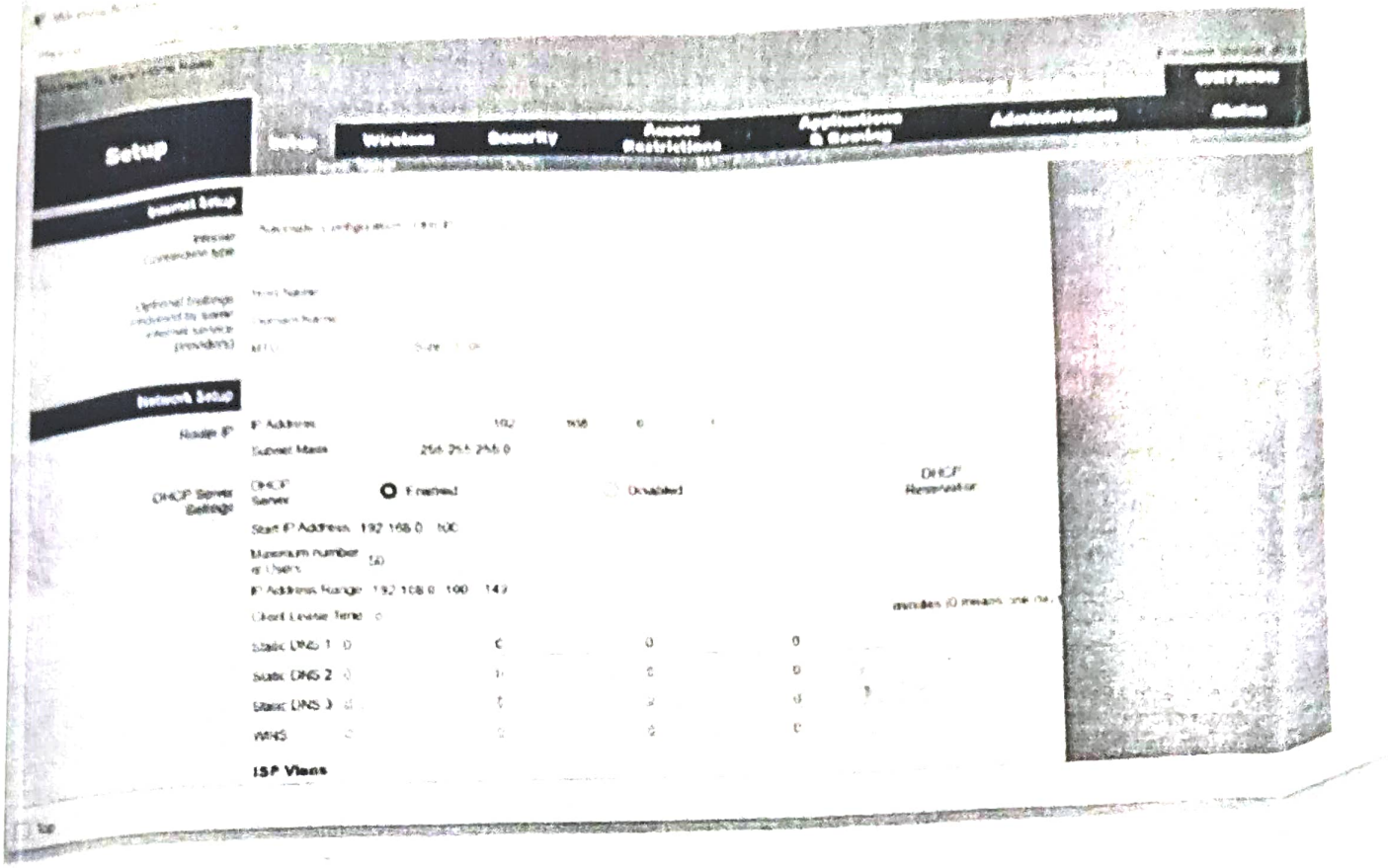
AIM:-

To design and configure an internetwork using a wireless router, DHCP server and Internet cloud in Cisco Packet Tracer.

OBSERVATION:-

1) A wireless router, PC, laptop, cable modem, Internet cloud and Cisco.com server are connected.
2) The wireless router is configured with the SSID HomeNetwork.
3) DHCP is enabled on the wireless router
4) The PC obtains its IPv4 address automatically through DHCP.
5) The laptop is configured with a wireless module and connected to HomeNetwork.
6) The internet cloud is connected to the cable modem.
7) The Cisco.com server is configured as a DHCP and DNS server.
8) The PC receives an address in the 192.168.0.x range.
9) Connectivity is verified using ping Cisco.com.

Home WiFi Router



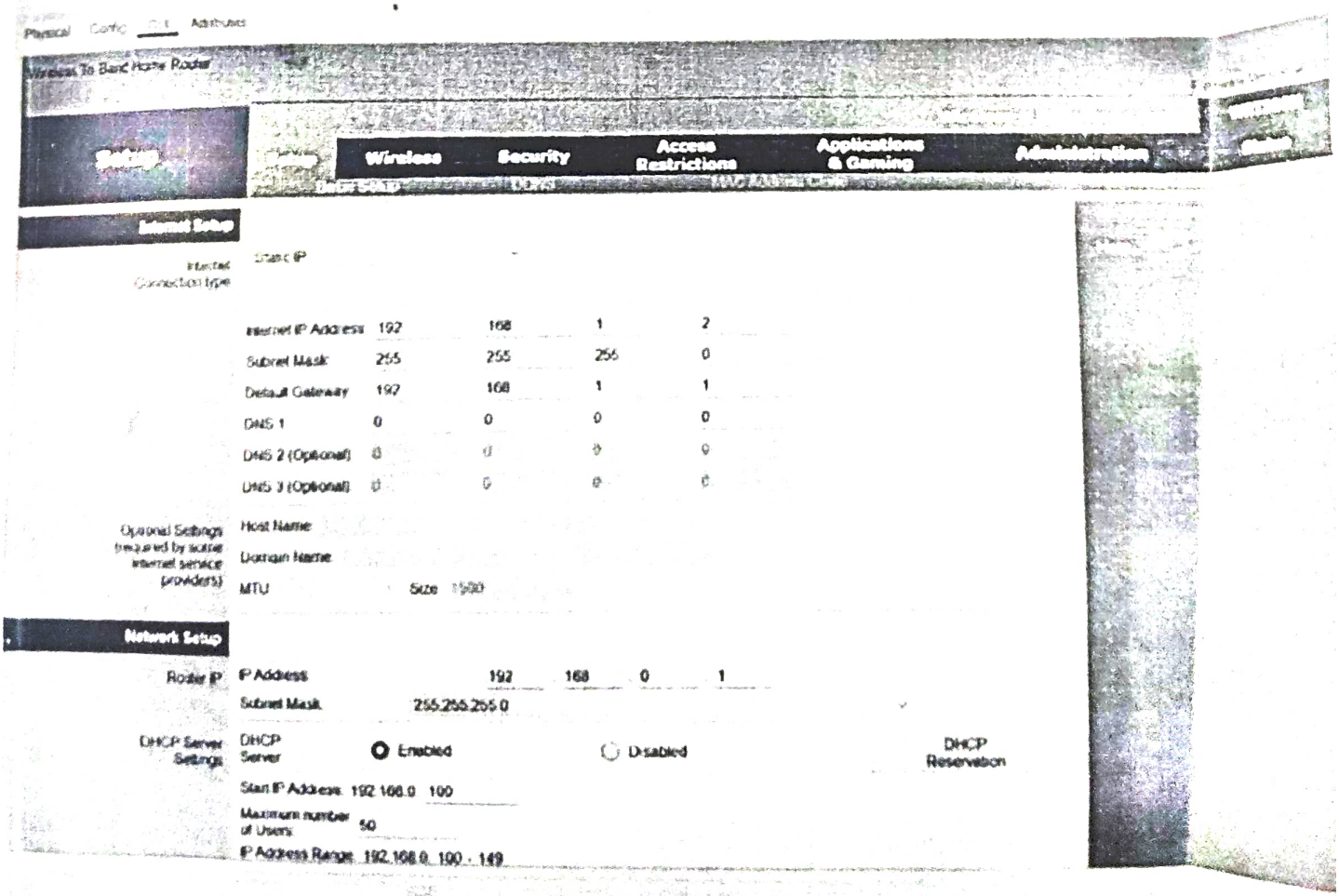
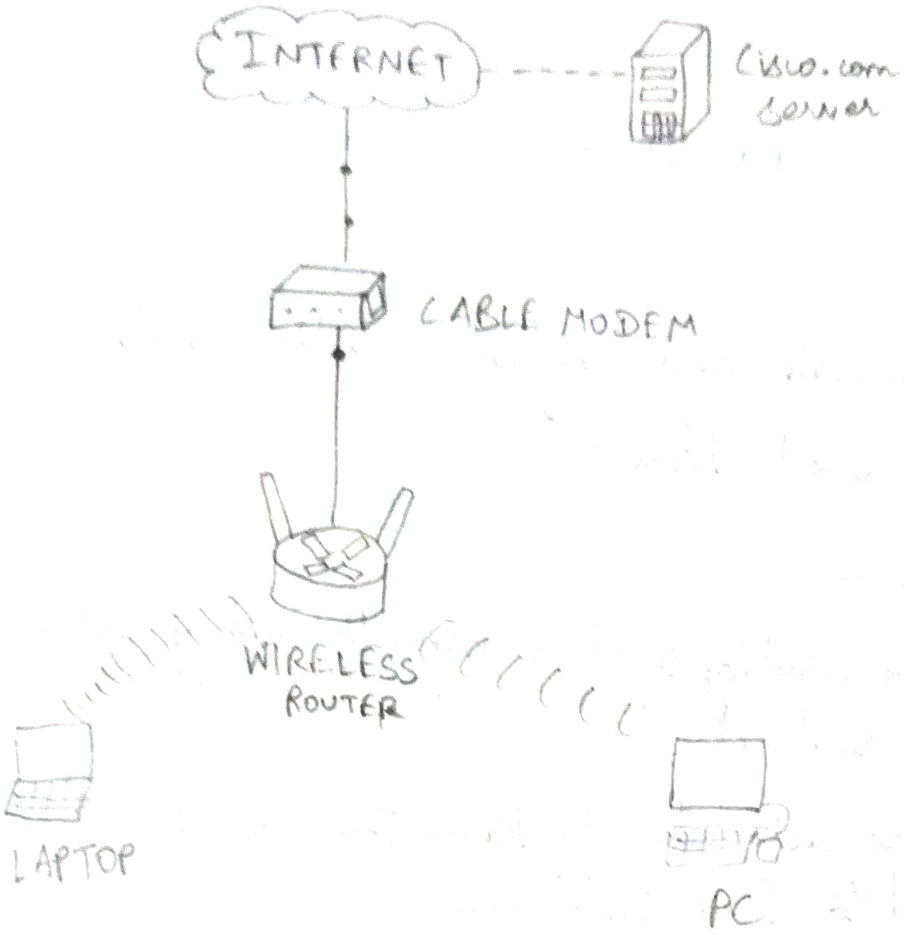


DIAGRAM:-



RESULT:-

Thus the wireless router, DHCP server and Internet cloud were successfully configured and connectivity to the Cisco.com server was verified.